

Day 6 - Ethernet LAN Switching (pt 2)

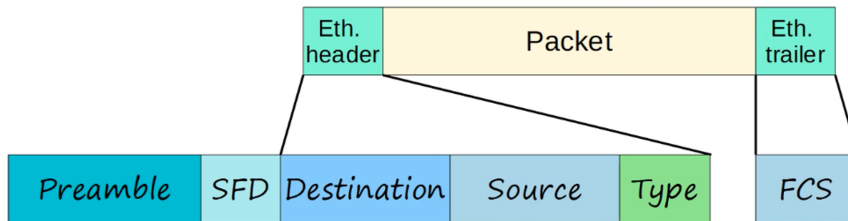
Thursday, August 20, 2026

3:28 PM

6. ETHERNET LAN SWITCHING : PART 2

An ETHERNET FRAME looks like:

Ethernet Header --- DATA (Packet) --- Ethernet Trailer



The Ethernet Header contains 5 Fields:

Preamble -- SFD -- Destination -- Source -- Type/Length 7 bytes -- 1 byte -- 6 bytes -- 6 bytes -- 2 bytes

Ethernet Trailer contains 1 Field:

FCS (Frame Check Sequence) = 4 bytes

- The PREAMBLE + SFD is not usually considered part of the ETHERNET HEADER. THEREFORE the size of the ETHERNET HEADER + TRAILER is 18 bytes

(6 + 6 + 2 + 4 bytes for the FRAME CHECK SEQUENCE)

The MINIMUM size for an ETHERNET FRAME (Header + Payload [PACKET] + Trailer) is 64 BYTES.

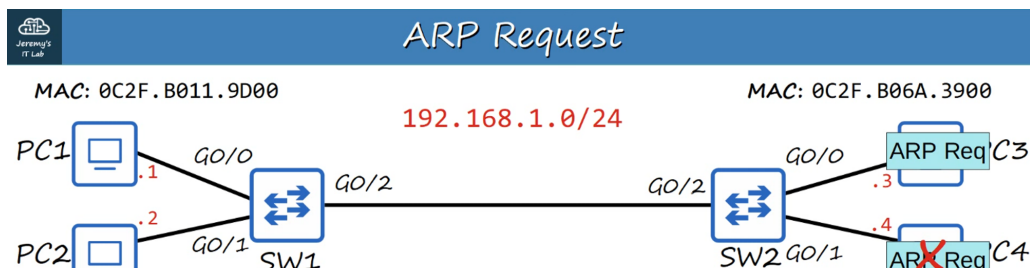
64 BYTES - 18 BYTES (Header + Trailer size) = 46 BYTES

THEREFORE the MINIMUM DATA PAYLOAD (PACKET) size is 46 BYTES!

IF the PAYLOAD is LESS than 46 BYTES then PADDING BYTES are added (padding bytes are a series of 0's) until it equals to 46 BYTES.

ARP

When a PC wants to send a packet to a destination, with a known IP address but an unknown MAC address, it first needs to send an ARP Request to learn that destination's MAC address.



MAC: 0C2F.B084.6200

SW1 MAC Address Table

MAC	Interface
.9D00	G0/0

ARP REQUEST
 Src IP: 192.168.1.1
 Dst IP: 192.168.1.3
 Src MAC: 0C2F.B011.9D00
 Dst MAC: FFFF.FFFF.FFFF

Src IP: 192.168.1.1
 Dst IP: 192.168.1.3
 Src MAC: .9D00
 Dst MAC: ???

MAC: 0C2F.B01E.0A00

SW2 MAC Address Table

MAC	Interface
.9D00	G0/2

- ARP stands for 'Address Resolution Protocol'.
- It is used to discover the Layer 2 address (MAC address) of a known Layer 3 address (IP address)
- Consists of two messages:
 - ARP REQUEST (Source message)
 - ARP REPLY (Destination message)
- ARP REQUEST is BROADCAST = sent to all hosts on network, except the one it received the request from.

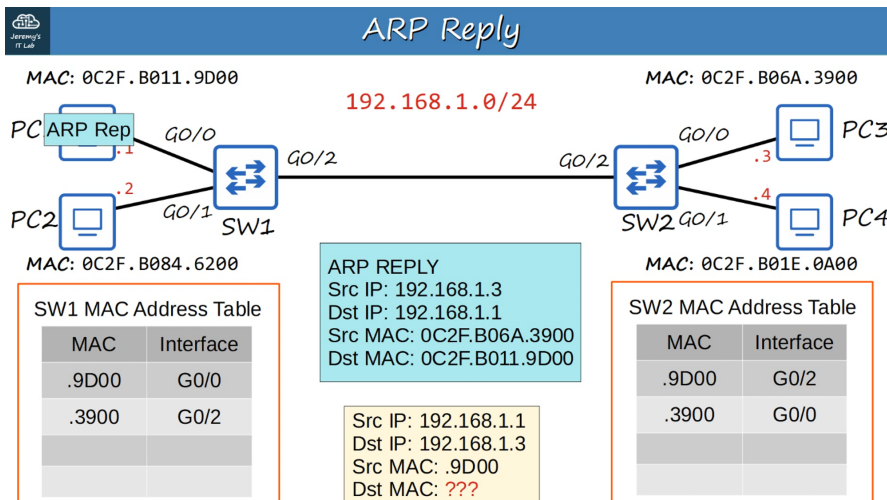
An ARP REQUEST frame has:

- Source IP Address
- Destination IP Address
- Source MAC address
- **BROADCAST MAC Address - FFFF.FFFF.FFFF**

An ARP REPLY frame has:

- Source IP Address
- Destination IP Address
- Source MAC address
- Destination MAC Address

ARP REPLY is a known UNICAST frame = Sent only to the host that sent the ARP REQUEST.



PING

- A network utility that is used to test reachability
- Measures round-trip time
- Uses two messages:
 - **ICMP Echo REQUEST**
 - **ICMP Echo REPLY**
- Is UNICAST

- Command to use ping:
 - ping

By Default, a CISCO IOS sends 5 ICMP requests/replies (Default size is 100-bytes)

- A period (.) is a failed ping
- An exclamation mark (!) is a successful ping

USEFUL CISCO IOS COMMANDS (from Privileged EXEC mode)

PC1# show arp // shows hosts ARP table



ARP Table

```

C:\Users\user>arp -a

Interface: 169.254.146.29 --- 0x0
Internet Address      Physical Address      Type
169.254.255.255       ff-ff-ff-ff-ff-ff    static
224.0.0.2             01-00-5e-00-00-02    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.0.167 --- 0xd
Internet Address      Physical Address      Type
192.168.0.1           98-da-c4-dd-a8-e4    dynamic
192.168.0.255         ff-ff-ff-ff-ff-ff    static
224.0.0.2             01-00-5e-00-00-02    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static
          
```

- Use `arp -a` to view the ARP table (Windows, macOS, Linux)
- Internet Address = IP address (Layer 3 address)
- Physical Address = MAC address (Layer 2 address)
- Type static = default entry
- Type dynamic = learned via ARP

SW1#show mac address-table // show the switches MAC table



MAC Address Table

```

SW1#show mac address-table
Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       0c2f.b011.9d00   DYNAMIC     Gi0/0
1       0c2f.b06a.3900   DYNAMIC     Gi0/2
Total Mac Addresses for this criterion: 2
SW1#
          
```

Will show:

Vlan --- MAC Address --- Type --- Ports(interfaces)

(Vlan = Virtual Local Area Network)



Clearing the MAC Address Table

```

SW1#show mac address-table
Mac Address Table
-----
Vlan    Mac Address      Type        Ports
          
```

clear mac address-table dynamic

```
Total Mac Addresses for this criterion: 2
SW1#clear mac address-table dynamic
SW1#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
SW1#
```

SW1# clear mac address-table dynamic

// clears the entire switches MAC table. // IF the optional MAC address is used, it will clear the SPECIFIC MAC address.

SW1 #clear mac address-table dynamic interface

// clears the MAC table entry of the Switch by it's INTERFACE name.